

A Ticket from Marginals to Joints: Coupled-Noise Distillation for One-Step Block Generation in Diffusion Language Models

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Abstract

Can a diffusion language model generate a coherent token block in one forward pass? Masked models already predict every position at once, but each prediction is the marginal distribution given the visible context, so the tokens can be mutually inconsistent and later steps revise those already committed. We introduce CONDOR (Coupled-Noise Distillation for One-Step Readout), trained from scratch to map different noise samples to different coherent blocks. Initially, random noise is not naturally paired with a target. Winner-take-all supervision lets different samples specialize, and self-distillation trains the one-pass output to match the refined coherent block. TinyStories experiments show diverse, coherent continuations over successive blocks, one forward pass each. Qualitative MNIST experiments show that the same approach can extend to multimodal generation, such as text-to-image and unconditional text-and-image generation.

1 Introduction

Standard autoregressive (AR) decoding generates one token per forward pass (39, 49). Conditioning each token on the preceding tokens supports coherent generation, but makes decoding sequential: a continuation of L tokens requires L forward passes. Subsequent work aims to preserve generation quality while improving decoding efficiency. Diffusion language models (dLLMs) extend the continuation one block after another, and improve the generation efficiency by decoding each new block in fewer than L refinement steps (Fig. 1). Within a block, two designs are used. Continuous-latent dLLMs denoise a continuous representation from Gaussian noise and decode it into tokens at the end. The continuous latent to be recovered must be obtained before or within the training procedure: some models take it from a frozen pretrained model (19), while others train a text VAE alongside the diffusion model (14) or learn the embeddings jointly (30). Masked dLLMs, including the LLaDA series (3, 36, 37), do not require this additional continuous latent. They operate directly on discrete tokens: each step fills several tokens in parallel until no mask remains in the current block. Parallel masked predictions add a further difficulty: each is based on the shared visible context, and tokens filled in the same step cannot see each other, so the predictions can be mutually inconsistent (13, 23). Later steps have to remove that inconsistency by revising the visible tokens (3, 20, 50). Both designs still produce one block over several steps. These limits raise our question: can a dLLM generate a coherent block in one forward pass? Especially, can it do so without a continuous latent obtained before or within training, and without a later revision of the visible tokens?

A masked dLLM already predicts every position of a block in one forward pass, yet the result need not be a coherent block. Under token-wise cross-entropy, the optimal prediction at each masked position is its marginal given the visible tokens (38), so sampling all positions at once draws from a product of marginals rather than from the joint distribution of the block. The obstacle is not the number of steps but the lack of a

per-sample variable that selects one joint mode. We give a masked dLLM such a variable by adding a Gaussian noise field ε to its mask embeddings. One noise sample (one *ticket*) should select one coherent block, and different tickets should select different blocks (Fig. 1d). Because noise is drawn independently of the data, the model is never told which block a ticket should map to, so this coupling must be learned. We train the dLLM from scratch to learn this coupling. CONDOR (**C**oupled-**N**oise **D**istillation for **O**ne-**S**tep **R**eadout) combines winner-take-all (WTA) supervision (16, 27, 35, 40) with self-distillation (Fig. 2). For a prefix and the one block about to be generated, we draw k noise samples. These samples then enter two branches. In the first, a ratio of positions in that block is masked and the remaining positions keep the ground truth. Each masked position adds its noise to the mask embedding before the model predicts. Only the prediction closest to the ground truth in cross-entropy is kept as the loss; the others are discarded. This symmetry breaking lets different noises learn different continuations. In the second, the whole block is masked and the k different noises are applied. For each noise ticket, we first iterate over several steps, filling only part of the mask each time, and obtain a relatively coherent sequence label. The one-step prediction is trained to match it, so one forward pass can produce that coherent block. WTA assigns each noise its own continuation, and self-distillation carries that continuation into the single pass.

Our contributions are as follows.

- We introduce a from-scratch model, CONDOR. By combining WTA supervision with self-distillation, it becomes the first diffusion language model that has the capability to generate one complete coherent token block per forward pass.
- We achieve coherent block-wise language generation. Each block is committed in one forward pass, and successive blocks remain a readable continuation.
- We achieve the same one-pass commitment in a multimodal setting, both for text-to-image and for unconditional generation of text and image, showing that the method generalizes beyond text and has potential for broader multimodal generation.

All experiments are conducted at small scale; our claims concern the mechanism rather than large-scale performance.



Figure 1: Committing a $B=8$ -token block. (a) AR, one token per pass. (b) Continuous-latent dLM, n denoising steps. (c) Masked dLLM, commit then edit. (d) Ours, one noise ticket per pass.

2 Related Work

Efficiency and joint coherence trade-off in autoregressive decoding. Autoregressive decoding spends one forward pass on each token, so the number of passes grows with the length of the continuation. Autoregressive models try to reduce that number of passes in several ways. Speculative decoding lets a lightweight model draft a few tokens; the large model checks them in one forward pass and keeps only the prefix it would itself have continued (29). Medusa predicts several tokens from one prefix in a single pass. The guesses cannot see each other, so the draft can conflict; verification then keeps the longest prefix consistent with the model’s own predictions (6). Jacobi decoding and Consistency LLMs fill a span of positions at once. The next round rewrites every position from the previous round’s left context, until each token is the greedy choice given that context (26, 44). These methods share a generate-then-self-verify procedure, and they still require multiple steps.

More efficient token generation with dLLMs. Diffusion language models reduce the number of passes in another way. They extend the continuation one block after another. A block of B tokens is generated in t iterative steps, with $t < B$. The two families differ in how that block is generated. Continuous-latent models denoise exactly as a standard diffusion model does, from Gaussian noise to a continuous representation (12, 14, 19, 30, 33, 55, 57). That continuous representation is built before or jointly with diffusion training, and it must be decodable back into tokens. It is taken from a frozen pretrained model (19), learned by a text VAE (14), or trained jointly as token embeddings (30). Because this denoising follows traditional diffusion, progressive distillation, consistency models, rectified flow, distribution matching, and MeanFlow can be expected to reduce the number of denoising steps here as well, even to one step (10, 31, 43, 48, 54). To our knowledge, continuous-latent language models have not yet adopted this further reduction. Masked models write discrete tokens and fill several positions in a step (1–3, 7, 32, 36, 38, 41, 46, 53). Parallel fills can be mutually inconsistent (23), so later work chooses which positions to commit (51), revises them, or turns them back into masks on subsequent passes (3, 11, 18, 20, 24, 45, 50, 56). Some methods introduce distillation from an already trained multi-step teacher to reduce the number of sampling steps (8, 17, 42). The sampling schedule becomes shorter, yet still requires multiple passes, except for DiMO, which has validated one-pass generation on images (59). CONDOR is trained from scratch, without a pretrained teacher, and is capable of committing each block in one forward pass for both text and images.

A necessary condition for one-step coherence. One step can commit several tokens in parallel. Those tokens form a coherent block only when the input and the output form a one-to-one or many-to-one pair: one input must select one coherent combination. Methods differ in how they form this pair. (i) VAE-like methods encode the target x into a high-dimensional latent z and decode z back into x , so z and x form the pair inside each training example (25). An earlier discrete-latent decoders (22), FlowSeq (34), LaNMT (47), VADD (52), and Cola DLM (14) are trained in this way. The pair fails when the decoder ignores the latent, as a conditional generator can ignore an input noise code, or when the posterior collapses onto the prior (4, 21, 58). (ii) Flow models form the pair from the initial noise. Rectified flow (31) and MeanFlow (10) carry one noise sample to one result in a single forward pass. ELF (19) is a dLLM trained with flow, so it has the same one-pass potential, but it does not take that step. Consistency distillation (48) and DMD (54) distill a diffusion model so that one noise sample yields one result in a single forward pass. DiMO (59) distills a masked diffusion model, taking a random token initialization as the noise, so that one initialization yields one result in a single forward pass. (iii) Under one condition, IWAE (5) uses different inputs and obtains different predictions of the same label. Every prediction remains in the loss. Instead of token-level cross-entropy, it multiplies the token probabilities into the sentence probability. This product represents the joint distribution of the continuation, rather than the per-position marginals. The weight on each input is a choice: the inputs can be averaged uniformly, or the weights can be tilted by the loss. (iv) Winner-take-all (WTA) losses (16, 27, 35, 40) also draw several inputs, but supervise only the lowest-loss one, so different inputs specialize to different outputs, as in Social GAN (15) and LoRA-MCL (28). CONDOR uses self-distillation as well as WTA. Each noise is self-distilled into a different label, so self-distillation also strengthens learning toward different labels.

3 Method

We first define f_θ , a block-wise dLLM based on a Transformer. A prefix c and a continuation block of B tokens are the input of f_θ . Attention inside a block is bidirectional. Across blocks, a token attends only to its own block and to earlier blocks (1). Each block to be predicted has ground truth x , and x_i is its token at position i . z is the current input of that block, formed from x and a mask position set A :

$$z_i = \begin{cases} [M] & i \in A, \\ x_i & i \notin A. \end{cases} \quad (1)$$

The masked fraction may be the whole block. Each visible token keeps its own embedding. Each mask embedding becomes a noised version,

$$\tilde{e}_i = e_{[M]} + \sigma \cdot \text{RMS}(e_{[M]}) \cdot \varepsilon_i, \quad i \in A, \quad (2)$$

where $\varepsilon \in \mathbb{R}^{B \times d}$ with d the embedding dimension, $e_{[M]}$ is the mask embedding, and σ scales the noise by $\text{RMS}(e_{[M]})$. Each row is drawn $\varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_d)$, or one vector $u \sim \mathcal{N}(0, I_d)$ is copied onto every $i \in A$. $\bar{x}_i$ is the vector of logits that f_θ predicts at position i :

$$\bar{x}_i = f_\theta(c, z, \varepsilon)_i, \quad i \in A. \quad (3)$$

CONDOR has two training tasks (Fig. 2). (i) Under a partial mask, k noises each produce a prediction. Every prediction is scored against the ground truth, but a gradient is sent only along the lowest-loss noise. The rest are dropped. (ii) Under a full mask, those k noises are self-distilled to k labels, one label per noise, and the one-pass prediction for a noise is trained to match its label.

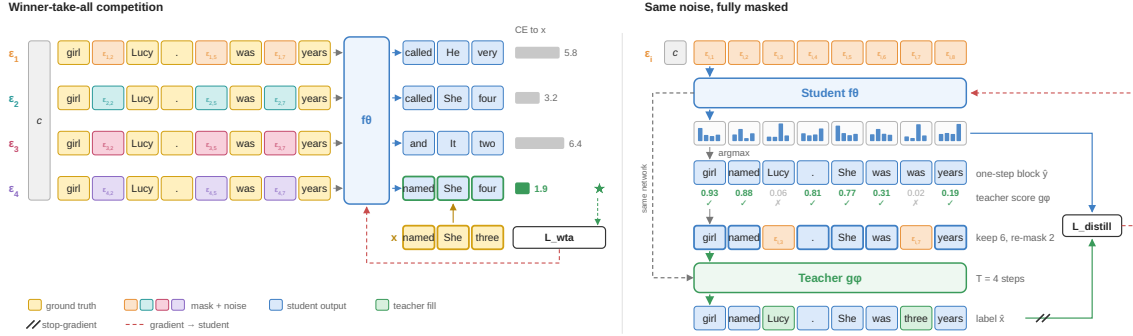


Figure 2: Training CONDOR. Left: the winner-take-all task. Right: self-distillation.

3.1 Winner-take-all (WTA) on a partial mask

On a fixed (c, z) we draw $k > 1$ noises $\varepsilon^{(j)} \in \mathbb{R}^{B \times d}$, $j = 1, \dots, k$. For noise j , the predicted logits are $\bar{x}_i^{(j)} = f_\theta(c, z, \varepsilon^{(j)})_i$. CE_j is the cross-entropy of these logits against the ground-truth tokens on the masked positions. j_{win} is the index of the noise with the lowest CE_j , and $\mathcal{L}_{\text{wta}}$ is that cross-entropy divided by $|A|$:

$$\text{CE}_j = \sum_{i \in A} \text{CE}(\bar{x}_i^{(j)}, x_i), \quad j_{\text{win}} = \arg \min_j \text{CE}_j, \quad \mathcal{L}_{\text{wta}} = \frac{1}{|A|} \text{CE}_{j_{\text{win}}}. \quad (4)$$

Only the winner sends a gradient. The other noises, the *losers*, are left free to stand for other continuations. Besides WTA, we ablate three other losses on these same noises: plain cross-entropy, IWAE, and IWAE with

weight.

$$\mathcal{L}_{\text{plain}} = \frac{1}{k|A|} \sum_{j=1}^k \text{CE}_j, \quad \mathcal{L}_{\text{iwae}} = -\frac{1}{|A|} \log \left(\frac{1}{k} \sum_{j=1}^k e^{-\text{CE}_j} \right). \quad (5)$$

IWAE with weight keeps the same joint mixture, but the mixture weights are not uniform. Let $c_j = \text{CE}_j/|A|$ be the mean token loss of draw j . Detached linear weights give the lowest-loss draw weight 1 and the highest-loss draw weight 0:

$$w_j = 1 - \frac{c_j - \min_m c_m}{\max_m c_m - \min_m c_m}, \quad \pi_j = \frac{w_j}{\sum_{m=1}^k w_m}, \quad \mathcal{L}_{\text{iwae,w}} = -\frac{1}{|A|} \log \left(\sum_{j=1}^k \pi_j e^{-\text{CE}_j} \right). \quad (6)$$

No gradient flows into w_j or π_j . Equal draw losses recover $\pi_j = 1/k$, so $\mathcal{L}_{\text{iwae,w}}$ equals $\mathcal{L}_{\text{iwae}}$. A zero weight removes that draw from the sum.

3.2 Self-distillation on a full mask

Self-distillation uses two networks, student f_θ and teacher g_ϕ . For a noise ε , the student and the teacher jointly form the distillation target $\hat{x}$. The parameters of g_ϕ can be the same as the current student, as in Fig. 2, or an exponential moving average of the student.

The block is fully masked, so $A = \{1, \dots, B\}$ and every position of the current input z is a mask. One forward of the student returns logits. These logits are greedily discretized into the one-pass tokens. The probability of token $\bar{y}_i$ on this forward is

$$\bar{x}_i = f_\theta(c, z, \varepsilon)_i, \quad \bar{y}_i = \arg \max_v f_\theta(c, z, \varepsilon)_{i,v}, \quad p_i = \text{softmax}(\bar{x}_i)_{\bar{y}_i}, \quad i = 1, \dots, B. \quad (7)$$

The one-pass tokens $\bar{y}$ are not necessarily a good continuation, so a kept ratio ρ of them stay, $\text{round}(\rho B)$ positions, and the rest are masked again and refilled. The kept positions may be chosen from the student’s p_i , the teacher’s p_i , or at random. When the student is the judge, p_i in Eq. (7) is sorted, the $\text{round}(\rho B)$ positions of the block with the highest p_i are kept, and the rest are remasked. When the teacher is the judge, position i is rescored by the teacher on an input that keeps the other tokens of $\bar{y}$ and replaces position i by the original noised mask $\tilde{e}_i$. This input is written $\bar{y}_{-i}$. g_ϕ returns logits, and p_i is the softmax probability of the proposed token:

$$p_i = \text{softmax}(g_\phi(c, \bar{y}_{-i}, \varepsilon)_i)_{\bar{y}_i}. \quad (8)$$

Then, by this p_i , the $\text{round}(\rho B)$ highest positions are kept and the rest are remasked. The teacher is then used to fill the remaining masks. In each step, the teacher commits the $(1 - \rho)B/T$ most confident argmax tokens until no mask remains, forming the filled block $\hat{x}$.

For each noise j , which is the same noise as in the WTA task, $\bar{x}^{(j)}$ is trained toward $\hat{x}^{(j)}$. The training loss is

$$\mathcal{L}_{\text{distill}} = \frac{1}{kB} \sum_{j=1}^k \sum_{i=1}^B \text{CE}(\bar{x}_i^{(j)}, \hat{x}_i^{(j)}), \quad \mathcal{L} = \mathcal{L}_{\text{wta}} + w(t) \mathcal{L}_{\text{distill}}, \quad (9)$$

with $w(t)$ increased early in training and then kept at 1.

3.3 Several blocks, and images

For block-wise generation inference, the model generates one block in a single forward pass, appends it to the prefix, and samples fresh noise for the next block. Each block is conditioned on the given prefix and all previously generated blocks. Repeating this procedure produces longer sequences while retaining one forward pass per block.

The same framework also supports multimodal generation, including text-to-image generation and unconditional generation of text and Images. In text-to-image generation, the text serves as the prefix for the image blocks. In unconditional generation, the model first generates text without a supplied text prefix, then generates the image conditioned on that text. Images are divided into patches, with each patch represented as a token. The model predicts continuous pixel values within each patch, and mean squared error replaces cross-entropy in both the winner-take-all and self-distillation objectives for image prediction. For image self-distillation, continuous patch predictions do not provide confidence scores for ranking patches. We therefore randomly select which patches of the one-pass prediction to remask, and randomly select which remaining masked positions to fill at each teacher step.

4 Experiments

4.1 Experimental setup

We evaluate CONDOR on four tasks spanning text and image generation, using a single forward pass per block throughout. *Single-block text continuation* generates an eight-token continuation conditioned on an eight-token prefix. *Multi-block text continuation* extends a text prompt with eight consecutive blocks of eight tokens each. *Word-to-image* generates a digit image conditioned on its English class name (*zero–nine*). *Unconditional* generates a digit word followed by its corresponding image.

Data. Both text tasks use TinyStoriesV2-GPT4 (9). For *Single-block text continuation*, each window is the first 16 GPT-2 tokens starting at a sentence boundary. For *Multi-block text continuation*, training samples begin at document boundaries. We tokenize the stories with the LLaDA tokenizer, group the tokens into blocks of eight, and pack them into sequences of up to 1024 tokens. Both image tasks use MNIST, with each 28×28 image partitioned into sixteen 7×7 patches.

Models and training. Table 1 summarizes core experimental settings. All CONDOR models are trained from scratch with AdamW at base learning rate 3×10^{-4} ; defaults are $k = 4$ and $\sigma = 0.5$. Additional implementation details are given in Appendix A.1.

Table 1: Core experimental settings for CONDOR. The image column applies to both image tasks.

	Single-block text continuation	Multi-block text continuation	Image generation
Architecture	Bidirectional Transformer	Block-causal Transformer	Bidirectional Transformer
Parameters	17.6M	0.5B	4.77M
Layers / hidden width / FFN width	6/256/1024	20/1024/2048	6/256/1024
Attention heads (query / key-value)	8/8	8/2	8/8
Total updates	150,000	16,000	50,000
Global batch size	512	32	512
Teacher	EMA	Current student	Current student
Full-mask winner treatment	Winner excluded	Winner GT	Winner GT
Keep ratio ρ / fill steps T	0.5/4	0.75/2	0.75/4
Retention	Confidence	Confidence	Random
Filling order	Confidence	Confidence	Random
Noise across positions	Independent	Shared within each block	Independent

The single-block basic setting is a comparison baseline, not the best configuration. In the row *Full-mask winner treatment*, *winner* is the noise already selected by partial-mask WTA. The partial-mask and full-mask tasks use the same set of noises, so that winner is handled inside the full-mask branch: *Winner excluded* drops

it from the full-mask loss, *Winner GT* trains it there against ground truth, and *Winner distilled* trains it there against the self-distillation result. The remaining noises use self-distillation results.

Evaluation. For fair comparisons, directly compared methods use matched evaluation examples and scoring protocols within their respective task settings. Each example provides an initial text prefix; each generated block contains 8 tokens. *Validity* is the fraction of blocks judged grammatically and contextually acceptable by GPT-6-Astra; natural truncation is allowed, earlier errors do not automatically invalidate the current block. For m generated blocks per example at a given block position, *Distinct* and D_m^{valid} are the mean numbers of distinct strings and distinct valid strings per example, respectively, including examples with no valid block; $d_{\text{all}} = \text{Distinct}/m$. *Uniqueness* is the ratio of distinct valid strings to valid blocks per example, averaged over examples with at least one valid block. *gen-PPL* is GPT-2-large perplexity of generated tokens given the conditioning prefix. MI_{16} measures whether a new noise sample changes the predictive distribution at the 8 token positions of the same prefix. At each position it is the entropy of the softmax averaged over 16 noise draws minus the mean entropy of those 16 softmaxes, and MI_{16} sums these eight values, in nats. The judging rubric appears in Appendix B.

For *Single-block text continuation*, we use the first 256 validation examples, each with an 8-token prefix, and generate 16 alternative 8-token blocks per example. *Validity*, D_{16}^{valid} , and *Uniqueness* use the first 32 examples (512 judged blocks); *Distinct*, *gen-PPL*, and MI_{16} use all 256 examples. For *Multi-block text continuation*, we use 4 evaluation examples and run generation 4 times per example, each run covering 8 successive 8-token blocks. *Free continuation* extends the initial prefix with earlier generated blocks; *GT prefix* uses the ground-truth prefix at each block position. Both settings use fresh noise for each block. We report *Validity*, d_{all} , and *Uniqueness* at each block position over $4 \times 4 = 16$ generated blocks, grouped by example.

To test how inference noise affects the continuation, we hold the trained model and the text prefixes fixed and change only the noise (Table 3). r_0 is the root-mean-square noise norm during training. The radius sweep keeps each noise direction and sets the norm to a multiple of r_0 . Automatic scores use 128×64 continuations, 64 draws for every one of 128 prefixes. D_{64} is the mean number of distinct strings among the 64 draws of one prefix. Judged *Validity* and D_{16}^{valid} use 32×16 continuations, 16 draws for every one of the first 32 prefixes. The angle sweep keeps the norm at r_0 and sets the angle to a reference noise on the same prefix. The reference output is the block generated from that reference noise. Within each reference group, pairwise token disagreement is the mean fraction of token positions that differ between output pairs, and reference change is the fraction of blocks that differ from the reference output in any token. *Conf.* is the mean model probability of the generated tokens.

4.2 Single-block text continuation

Table 2 compares single-block text continuation methods. Following DiMO (59), the text run uses the Plain CE model as the frozen teacher and distills it into a one-step generator. This adaptation produces no valid outputs in the judged evaluation (Appendix A.2). The results support three observations. (1) Without self-distillation, Plain CE, IWAE, IWAE with weight, and WTA all underperform CONDOR in one-pass *Validity* and valid-output diversity. IWAE with weight replaces the uniform $1/k$ mixture in Eq. (5) by the detached loss weights in Eq. (6). Its one-pass *Validity* is 6.4% and D_{16}^{valid} is 0.78. The 30k-update CONDOR row aligns training time with those 150k-update baselines: self-distillation slows each update, so the schedule stops at 30k updates (2.9 GPU-hours, within the 5.3 GPU-hours of WTA at $k = 4$ and the 8.3 GPU-hours at $k = 16$; Appendix A). On this budget, one-pass *Validity* is 60.4% and D_{16}^{valid} is 7.38, above Plain CE, IWAE, IWAE with weight, and one-pass WTA. The single-block case study qualitatively supports this finding for WTA versus CONDOR (Table 5, top). (2) No output collapse is observed when self-distillation targets are generated by the current student, an EMA of the student, or a frozen WTA model. (3) Ablations of the keep ratio, token-retention rule, competition size, and full-mask winner treatment identify the best tested one-pass *Validity* of 93.9%: current student as teacher, $\rho = 0.75$, noise shared across tokens, and *Winner distilled*.

Holding the trained model and the text prefixes fixed, we change only the inference noise (Table 3). Under that probe, a larger angle from the reference noise increases the fraction of outputs that differ from the reference

Table 2: Single-block text continuation results. * assigns the same noise vector to every masked position in the block, in both training and inference. Other rows draw an independent noise vector at each masked position. The 30k-update CONDOR row is the wall-clock alignment with the no-self-distillation runs.

Method / variant	Validity						
	NFE	(%) $\uparrow$	$D_{16}^{\text{valid}} \uparrow$	Uniqueness $\uparrow$	Distinct $\uparrow$	gen-PPL $\downarrow$	MI ₁₆
<i>No self-distillation</i>							
Plain CE	1	9.0	0.25	0.46	2.6	285.0	0.02
Plain CE*	1	6.4	0.12	0.14	2.2	316.7	0.01
IWAE	1	4.5	0.66	0.95	13.7	249.6	2.17
IWAE with weight	1	6.4	0.78	0.86	14.1	241.7	2.27
IWAE*	1	6.2	0.72	0.84	12.7	243.5	2.15
WTA, $k = 4$	1	9.2	1.19	0.90	14.3	242.1	2.38
WTA, $k = 4^*$	1	5.3	0.72	0.92	13.9	242.4	2.37
WTA, $k = 16$	1	18.4	2.72	0.96	15.5	138.4	6.05
WTA, $k = 4$, 8 passes	8	97.3	14.91	0.96	15.1	20.7	—
DiMO	1	0.0	0.00	—	16.0	2702.5	13.45
CONDOR	1	81.6	11.31	0.87	13.6	24.5	7.66
CONDOR*	1	82.2	9.69	0.75	11.9	24.1	6.95
CONDOR, 30k updates	1	60.4	7.38	0.80	12.3	36.2	5.62
Keep ratio $\rho = 0.25$	1	76.2	10.97	0.91	15.0	34.9	9.60
Keep ratio $\rho = 0.75$	1	88.3	11.50	0.82	12.9	20.4	6.94
Keep ratio $\rho = 0.75^*$	1	76.4	8.88	0.75	11.7	27.8	8.13
$k = 8$, global batch 256	1	76.6	11.81	0.96	15.4	29.0	9.80
$k = 16$, global batch 128	1	74.0	11.72	0.99	15.9	41.4	11.41
Independent target noise ε'	1	90.0	1.38	0.10	1.7	15.6	0.37
Retention: student confidence	1	81.8	11.84	0.90	14.5	28.0	8.34
Winner distilled	1	83.6	11.94	0.90	13.9	22.4	8.11
Winner distilled*	1	81.1	10.06	0.78	12.2	24.9	8.08
Teacher: frozen WTA	1	90.8	11.31	0.78	11.6	21.9	4.81
Teacher: current student	1	82.2	11.16	0.85	14.3	24.9	8.74
Teacher: current student*	1	85.2	9.91	0.74	11.9	25.3	7.27
Teacher: current student, $\rho = 0.75$, Winner GT	1	79.3	9.78	0.79	12.6	22.7	4.69
Teacher: current student, $\rho = 0.75$, Winner GT*	1	80.3	8.22	0.67	11.0	37.3	4.94
Teacher: current student, $\rho = 0.75$, Winner excluded*	1	84.4	9.56	0.72	10.6	34.4	5.65
Teacher: current student, $\rho = 0.75$, Winner distilled	1	79.1	8.97	0.74	11.5	26.9	4.17
Teacher: current student, $\rho = 0.75$, Winner distilled*	1	93.9	11.62	0.77	12.3	26.3	6.02

output, while gen-PPL stays stable (Table 3). A larger norm produces more distinct outputs, but an excessive norm reduces Validity. Both the direction and the magnitude of the noise affect the continuation.

4.3 Block-wise text generation

We evaluate whether the 0.5B CONDOR model can extend generation across 8 successive 8-token blocks (Table 4). Uniqueness increases across blocks in both Free continuation and GT prefix. Free-continuation Validity declines over the eight blocks, while GT-prefix Validity is largely retained.

The stories in Table 5 (bottom) show that Free continuation with the same 0.5B checkpoint can extend a prefix through successive blocks into a longer text. Each new block is conditioned on the previously accepted generated blocks; unsatisfactory blocks are regenerated with fresh noise before continuation proceeds.

Table 3: Inference-noise norm and direction for the default seed-0 student.

<i>Fixed radius</i>						<i>Fixed angle, norm r_0</i>					
Radius	Automatic: 128×64			Judged: 32×16			Within each reference cone				
	D_{64}	Conf. (%)	gen-PPL $\downarrow$	Validity (%)	D_{16}^{valid}	Angle	D_{64}	Pairwise token disagreement (%)	Reference change (%)	Conf. (%)	gen-PPL $\downarrow$
Original Gaussian	43.55	75.9	25.23	83.2	11.34	0°	1.00	0.0	0.0	75.2	24.82
$0 \times r_0$	1.00	71.2	21.67	71.9	0.72	5°	4.11	8.9	27.3	75.3	24.76
$0.25 \times r_0$	27.48	73.6	22.01	83.2	8.91	15°	11.48	23.4	54.2	75.6	24.36
$0.5 \times r_0$	36.95	75.4	23.46	84.2	10.59	30°	23.03	39.7	75.2	75.7	24.46
$0.75 \times r_0$	41.10	75.9	24.50	84.6	11.25	45°	32.23	50.8	86.0	75.9	24.74
$1 \times r_0$	43.55	75.9	25.22	82.8	11.28	60°	38.75	58.0	92.0	76.0	24.59
$1.25 \times r_0$	46.55	75.3	26.16	84.0	11.75	90°	43.80	62.8	97.2	76.1	24.79
$1.5 \times r_0$	49.53	74.1	27.84	79.5	11.28	120°	38.72	57.3	99.0	76.0	25.03
$2 \times r_0$	57.52	66.7	42.80	61.7	9.53	150°	22.88	39.7	99.6	76.0	24.99
$3 \times r_0$	64.00	41.5	1142.27	0.8	0.12	180°	1.00	0.0	99.8	75.8	25.44

Table 4: Block-wise generation with the 0.5B student at step 16,000. All values are percentages, based on 4 evaluation examples with 4 generation runs per example. Validity scores the current block, not the entire continuation.

Block	Free continuation			GT prefix		
	Validity	Uniqueness	d_{all}	Validity	Uniqueness	d_{all}
1	100.00	25.00	25.00	100.00	25.00	25.00
2	87.50	50.00	50.00	100.00	43.75	43.75
3	87.50	56.25	50.00	93.75	54.17	56.25
4	87.50	81.25	81.25	100.00	50.00	50.00
5	75.00	91.67	93.75	68.75	50.00	62.50
6	87.50	91.67	93.75	93.75	45.83	50.00
7	81.25	91.67	93.75	62.50	52.78	50.00
8	68.75	87.50	93.75	93.75	56.25	56.25

4.4 Multimodal generation

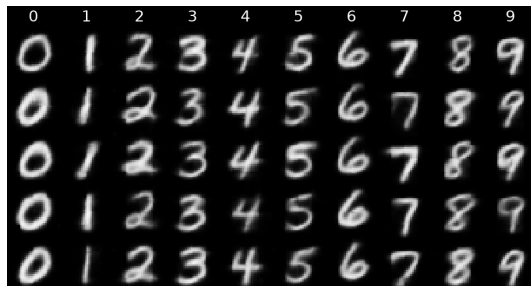
We demonstrate the same block-wise generation mechanism beyond text in two multimodal settings (Fig. 3). In *text-to-image generation*, an English digit word conditions an image block predicted in one forward pass. In *unconditional text-and-image generation*, the model first generates a digit word and then an image conditioned on that word, using one forward pass per block. These MNIST examples provide a qualitative demonstration of multimodal generation.

Table 5: Qualitative text-generation examples. Top: one-pass continuations of the same eight-token prefix; underlining marks errors. Bottom: selected stories from the 0.5B student at step 16,000. Each attempt uses one forward pass; these stories are not unfiltered rollouts.

WTA	CONDOR
Prefix: <i>At the market, Lily saw many fruits</i> <i>. She were to apples,,,</i> <i>. She saw to apples,,,</i>	<i>inside. She said, "Wow,</i> <i>on the ground. She said, "</i>
Multi-block text continuation. <i>One day, a little girl named Sue went to the park. She saw a big tree and wanted to climb it. Sue started to climb the tree, but she was very small. Sue saw a big bird in the tree. The bird said, "Hi, Sue! Do you need help?" Sue said, "Yes, I want to climb the tree." The bird said, "I can help you, Sue." Sue and the bird climbed the tree. They were happy and said, "Thank you, bird!" The bird said, "You're welcome, Sue." Sue and the bird became good friends. They played in the park every day.</i>	
Multi-block text continuation. <i>Sara and Tom are friends. They like to play with toys in Sara's room. Sara has many toys. She has dolls, cars, blocks and books. Tom likes to play with toys. One day, Sara says, "Tom, let's play together." Tom says, "OK, Sara, I will play with you." Sara and Tom play with the toys together. They have a lot of fun. They are happy.</i>	



(a) Text-to-image generation (step 18,000). Columns are zero-nine; each sample row uses the same noise across words.



(b) Unconditional text-and-image generation (step 18,500). The digit word is generated before the image.

Figure 3: Qualitative multimodal generation on MNIST, with one forward pass per generated block.

5 Conclusion

We introduced CONDOR, a noise-conditioned masked diffusion model trained from scratch with winner-take-all supervision and same-noise self-distillation. The method learns to map different noise samples to different continuations, predicting a complete block in one forward pass. Small-scale experiments demonstrate single-block and block-wise text generation, as well as text-to-image and unconditional text-and-image generation.

6 Limitations

The experiments here study TinyStories continuation and a qualitative MNIST demonstration. Larger models, more complex and diverse datasets, reliable long-form generation, and broader generalization are natural directions for follow-up work.

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Supplementary Material

Appendix A provides architecture, training schedules, the DiMO adaptation, and training compute. Appendix B defines the judging rubric. Appendix C reports additional experiments and examples.

A Additional experimental details

A.1 Architecture and training details

Table S1 complements the core settings in Table 1. The single-block model uses pre-norm layers, learned positional embeddings, and tied input/output token embeddings. Its EMA teacher has decay 0.9999, is initialized from the student, and is updated after each optimizer step.

Table S1: Optimizer details for CONDOR. The image column applies to both image tasks.

	Single-block text	Multi-block text	Images
AdamW (β_1, β_2)	(0.9, 0.999)	(0.9, 0.95)	(0.9, 0.999)
Weight decay	0.01	0.1	0.01
Warm-up updates W	2000	800	2500
Final learning rate	0	3×10^{-5}	0

Learning-rate and distillation schedules. Let S denote total updates and $\eta_0 = 3 \times 10^{-4}$. Single-block training uses $\eta_t = \eta_0 \min(1, t/W)[1 + \cos(\pi t/S)]/2$ at update t , with distillation weight $w(t) = 0.1 + 0.9 \min(t/80,000, 1)$. The 30k-update runs set $S = 30,000$ and shorten the distillation ramp to 16,000 updates, so wall-clock time stays with the 150k-update runs that do not use self-distillation. Multi-block and image runs use linear warm-up over W updates followed by cosine decay to the final rate in Table S1. Their distillation weights are $w(t) = \min(1, \max(0, (t - 3200)/3200))$ and $w(t) = 0.1 + 0.9 \min(t/20,000, 1)$, respectively. The multi-block full-mask loss is normalized by k ; for images, both the full-image ground-truth term and the refilled-pixel distillation term are normalized by k .

Training compute. All single-block runs use one A100. Default CONDOR takes 14.1 GPU-hours, the $k = 4$ WTA baseline 5.3 (8.3 at $k = 16$), and each 30k-update run 2.9. DiMO takes about 2 GPU-hours per configuration.

A.2 Baseline training

Plain CE, IWAE, and WTA. These runs use the same base learning rate and weight decay as the single-block basic setting, with uniform mask sampling unless specified otherwise. Their learning rate uses 2000 linear warm-up updates followed by cosine decay to zero. IWAE with weight uses this same schedule and $k = 4$. Its objective is Eq. (6).

DiMO. DiMO originally distills a pretrained teacher on discrete images. The text adaptation follows that protocol. The 150k-update Plain CE checkpoint is the frozen teacher. Student, teacher, and fake model are initialized from this checkpoint. The four adaptations cross fully masked or hybrid inputs with learning rate 10^{-5} (embeddings frozen) or 10^{-4} (all parameters trainable), each for 50k additional updates at batch size 512. Table 2 reports the hybrid-input adaptation at learning rate 10^{-5} . Each update samples a student continuation and remasks a random subset. The teacher-fake forward KL at masked positions updates the student. A fresh student sample trains the fake model with masked cross-entropy. Fully masked inputs use $\sigma = 0.5$ noise; hybrid inputs use 60% masks and 40% uniformly sampled tokens, with embeddings perturbed

as $\sqrt{1 - 0.1^2} e + 0.1 \text{RMS}(E)\epsilon$. All four adaptations use AdamW with zero weight decay and the single-block learning-rate formula above with $S = 50,000$ and $W = 500$. Sampling temperature is 1; each student update is paired with one fake-model update using label smoothing 0.1.

A.3 Ablation variants

The independent-target-noise run builds the distillation target under a fresh ϵ' and trains the student under the original ϵ .

B Evaluation protocols and metrics

B.1 Frozen-rubric text-validity re-review

Criterion. A continuation is judged together with its visible prefix. Natural truncation is allowed. Internal grammar errors, merged words, token loops, control tokens, and punctuation degeneration fail. A close-quote can pass when the visible text already reads as the end of speech. An extra close-quote or a nesting the visible text already breaks still fails. Boundary items are concrete ambiguities, such as a name switch. The reported *mixed* policy uses each item’s recorded decision. The judge is a language model that had seen earlier rubric discussion, so this is not an independent human review. Labels were locked before aggregation.

C Additional experiments and examples

C.1 Noise geometry

Tables 3 and S2 probe the default seed-0 student at 150k updates, with no retraining. A radius- αr_0 input uses a unit direction u in the $D = 2048$ completion field, $\delta = \sigma \text{RMS}(e_{[M]}) \alpha \sqrt{D} u$, so $\|\delta\|_2 = \alpha r_0$. On this checkpoint $\text{RMS}(e_{[M]}) = 0.720720$ and $r_0 = 16.308026$. An angle- θ input has the same norm and angle θ to a reference direction, with the tangent uniform on the orthogonal complement. The 0° and 180° cases are a single direction.

Table S2: Unedited continuations of *They would swim, splash, and pretend* at norm r_0 . Panel A follows one tangent; panel B uses other directions at 30° .

Angle	Tangent	Generated continuation	Conf. (%)	$\log P$
<i>A. One paired rotation trajectory</i>				
0°	—	to play. The fish said, "	78.4	−2.163
30°	v_0	to play. Fin said, "Yes	89.2	−1.063
90°	v_0	to swim.\nOne day, the	91.0	−0.961
180°	—	to be friends.\nOne day,	86.5	−1.259
<i>B. Multiple directions with the same angle to the same reference</i>				
30°	v_1	together. The fish said, "Okay	86.1	−1.472
30°	v_2	to play. The fish said, "	92.6	−0.624
30°	v_4	to be a fish. One day,	61.2	−4.808
30°	v_6	to play.. fish said, "	74.8	−2.545